CUSTOMER MILESTONE BRIEF

MS #07 - Autonomous Factory Run Engine

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.7.0, 0.7.1
Date	2026-08-11 to 2026-08-11
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #7 - Autonomous Factory Run Engine converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Introduces controlled planner, builder, and verifier automation with deterministic policy remaining authoritative.

Why this matters to a customer

Introduces controlled planner, builder, and verifier automation with deterministic policy remaining authoritative.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added a deterministic autonomous-factory controller around replaceable planner, builder, and verifier agents; workers propose artifacts while deterministic policy remains authoritative.
- Added versioned work-order, agent-artifact, and run-receipt contracts with strict path, attempt, file-count, patch-size, timeout, audience, and network-intent policies.
- Added copy-on-run workspaces, exact bounded edit application, command-array acceptance gates, content-addressed artifacts, and an append-only hash-chained event ledger.
- Added implementer/verifier separation so private gate output and diagnoses are withheld from the builder while public failure envelopes remain useful for repair.
- Added local reference agents and an optional OpenAI Responses API adapter using strict structured output, disabled response storage, server-side credentials, and interchangeable model selection.
- Added an offline five-fault INTCALC mutation gauntlet that proves every defect is rejected before repair and reports autonomous repairs and false acceptances separately.
- Added a Factory control room to the graph explorer for run status, gates, state transitions, changed paths, receipt identity, and audience-filtered event inspection.

- Expanded modern source indexing so factory, graph, oracle, benchmark, and test Python files are searchable source nodes with content-addressed evidence in the canonical graph.
- Added factory contract, path-boundary, ledger-tamper, privacy, provider, benchmark, HTTP, and UI tests. Mainframe equivalence remains explicitly unclaimed until runtime evidence is connected.
- Added shared macOS/Linux Python admission that automatically selects Python 3.11 or newer, supports an explicit LIGHTYEAR_PYTHON override, and rejects Apple's bundled Python 3.9 before a run can be misclassified as a semantic failure.
- Made the offline benchmark candidate postpone annotation evaluation as a defensive import guard.
- Added collision-safe, path-opaque run keys so repeated benchmark collections with the same human-readable run IDs remain independently addressable in the Factory control room.
- Added supported/unsupported runtime-admission and repeated-collection regression tests.

How it differs from earlier work

MS #7 builds on MS #6 (Relationship and Source Evidence Intelligence). The earlier milestone established its own bounded capability; this milestone adds autonomous factory run engine while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use autonomous factory run engine as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- Added factory contract, path-boundary, ledger-tamper, privacy, provider, benchmark, HTTP, and UI tests. Mainframe equivalence remains explicitly unclaimed until runtime evidence is connected.

Source of record

- CHANGELOG.md release section(s): 0.7.0, 0.7.1
- README.md product and operating guidance

- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity